

Unit-3 : Functions

Functions Topic Table

No.	Topic	Main Concepts	Important Points	Exam Focus
1	Introduction to Functions	Meaning of function	Relation where every input has exactly one output	Definition based questions
2	Domain	Set of input values	Values allowed in function	Find domain
3	Codomain	Possible output set	Declared output set	Difference from range
4	Range	Actual output values	Subset of codomain	Find range
5	Representation of Functions	Ordered pairs, mapping, formula	Different representation methods	Drawing mapping diagrams
6	Types of Functions	One-One, Many-One, Onto, Into	Classification of functions	Identify function type
7	Injective Function	One-One function	Different inputs $\rightarrow$ different outputs	Proof and examples
8	Surjective Function	Onto function	Every codomain element mapped	Range = codomain
9	Bijjective Function	One-One + Onto	Important for inverse	High-weight topic
10	Identity Function	$f(x)=x$	Every element maps to itself	Simple definitions
11	Constant Function	Same output for all inputs	Range has single element	Easy examples
12	Composition of Functions	$(f \circ g)(x)=f(g(x))$	Order matters	Numerical problems
13	Associative Property	$f \circ (g \circ h) = (f \circ g) \circ h$	Composition is associative	Theory question
14	Inverse Function	$f^{-1}(x)$	Exists only for bijection	Important theorem
15	Finding Inverse	Replace x and y method	Solve for y	Problem solving
16	Verification of Inverse	$f(f^{-1}(x))=x$	Identity verification	Practical questions
17	Applications of Functions	Mapping systems, coding	Computer science usage	Short notes

Algebraic Structures Topic Table

No.	Topic	Definition Idea	Key Property	Example
1	Binary Operation	Operation on set	Closure	$a*b$
2	Algebraic Structure	Set + operation	Mathematical system	$(S, *)$
3	Closure Property	Result stays in set	$a*b \in S$	Integers under +
4	Associative Property	Grouping unchanged	$(ab)c=a(bc)$	Addition
5	Identity Element	Neutral element	$a*e=a$	0 in addition
6	Inverse Element	Opposite element	$a*a^{-1}=e$	-a

7	Commutative Property	Order unchanged	$ab=ba$	Addition
8	Semigroup	Closure + Associative	Minimum structure	Natural numbers under +
9	Monoid	Semigroup + Identity	Has identity	Integers under +
10	Group	Monoid + Inverse	Main topic	Integers under +
11	Abelian Group	Commutative group	$ab=ba$	Real numbers under +
12	Finite Group	Limited elements	Order finite	Z_4
13	Infinite Group	Unlimited elements	Infinite order	Integers
14	Order of Group	Number of elements		G
15	Cyclic Group	Generated by one element	Repeated operation	$\langle a \rangle$
16	Generator	Creates entire group	Important concept	1 generates Z_5
17	Subgroup	Smaller group inside group	Must satisfy group rules	$H \subseteq G$
18	Proper Subgroup	Not equal to whole group	$H \neq G$	Example based
19	Trivial Subgroup	$\{e\}$ and G	Basic theorem	Identity only
20	Normal Subgroup	$gH=Hg$	Important for quotient	$N \triangleleft G$
21	Cosets	Left and right partitions	gH or Hg	Lagrange theorem
22	Homomorphism	Structure-preserving map	$f(a*b)=f(a)f(b)$	Mapping groups
23	Kernel	Maps to identity	Important theorem	$\ker(f)$
24	Isomorphism	One-One homomorphism	Structures equivalent	Group equivalence
25	Lagrange's Theorem	Subgroup order divides group order	Core theorem	Very important
26	Congruence Relation	Modular relation	$a \equiv b \pmod{n}$	Modular arithmetic
27	Quotient Structure	Group formed using cosets	G/N	Advanced concept
28	Ring	Two operations $(+, \times)$	Addition group + multiplication	Integers
29	Commutative Ring	Multiplication commutative	$ab=ba$	Integers
30	Integral Domain	No zero divisors	$ab=0 \Rightarrow a=0 \text{ or } b=0$	Integers
31	Field	Every nonzero element invertible	Division possible	Rational numbers

Functions — Basics

A function is one of the most important concepts in Discrete Mathematics.

1. Introduction to Functions

Definition: A **function** is a special type of relation where:

Every input has exactly one output.

If input is repeated with different outputs, then it is **not a function**.

Mathematical Representation

If function name is (f),

$$f : A \rightarrow B$$

Where:

Symbol	Meaning
(A)	Domain (input set)
(B)	Codomain (possible outputs)
(f)	Function

Basic Example

$$f(x) = 2x + 1$$

Input:

x	Output (f(x))
1	3
2	5
3	7

Each input gives exactly one output.

Therefore, it is a function.

Real-Life Analogy

Input	Function	Output
Student Roll No	Search System	Student Name
Employee ID	Company Database	Employee Record
Number	Multiply by 2	Result

Important Rule

Valid Function

Input	Output
1	A
2	B
3	C

Each input → one output only.

Not a Function

Input	Output
1	A
1	B

Same input has two outputs.

Therefore NOT a function.

2. Domain

Definition: The **domain** is the set of all input values allowed in a function.

Example

$$f(x) = x^2$$

If:

$$A = \{1,2,3,4\}$$

Then:

x	(f(x))
1	1
2	4
3	9
4	16

Domain:

$$\{1,2,3,4\}$$

Important Notes

Point	Explanation
Domain = Inputs	Starting values
Usually represented by A	$\ln (f:A \rightarrow B)$
No invalid inputs allowed	Example: denominator $\neq 0$

Example with Restriction

$$f(x) = \frac{1}{x}$$

Domain cannot contain:

$$x=0$$

Because division by zero is undefined.

3. Codomain

Definition: The **codomain** is the set of all possible outputs declared for the function.

Example

$$f:A \rightarrow B$$

Where:

$$A=\{1,2,3\}$$

$$B=\{1,4,9,16,25\}$$

and

$$f(x) = x^2$$

Outputs obtained:

x	(f(x))
1	1
2	4
3	9

Codomain is still:

$$\{1,4,9,16,25\}$$

Because codomain means all possible declared outputs.

Key Point

Term	Meaning
Codomain	Possible outputs

Range	Actual outputs
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Students commonly confuse these two.

4. Range

Definition: The **range** is the set of actual outputs produced by the function.

Example

Using previous example:

$$f(x) = x^2$$

Domain:

{1,2,3}

Outputs:

{1,4,9}

Therefore Range:

{1,4,9}

Difference Between Codomain and Range

Codomain	Range
Possible outputs	Actual outputs
Declared set	Obtained set
May contain extra elements	Never extra elements

Visual Understanding

Codomain = {1,4,9,16,25}

Range = {1,4,9}

16 and 25 are possible but not actually produced.

5. Representation of Functions

Functions can be represented in multiple ways.

A. Ordered Pair Representation

Example

$f = \{(1,2), (2,4), (3,6)\}$

Meaning:

Input	Output
1	2
2	4
3	6

Important Rule

First element should not repeat with different outputs.

Valid

$\{(1,2), (2,4), (3,6)\}$

Invalid

$\{(1,2), (1,5)\}$

Because input 1 has two outputs.

B. Mapping Diagram

Example

$$A = \{1, 2, 3\}$$

$$B = \{2, 4, 6\}$$

$$1 \rightarrow 2$$

$$2 \rightarrow 4$$

$$3 \rightarrow 6$$

This visually shows function mapping.

C. Formula Representation

Example

$$f(x) = 2x$$

Find outputs using formula.

x	f(x)
1	2
2	4
3	6

Important GTU Exam Questions

Question Type	Example
Define function	Theory
Find domain/range	Numerical
Difference between range and codomain	Theory
Check whether relation is function	Conceptual
Write ordered pair representation	Practical
Draw mapping diagram	Diagram-based

Most Important Concept

A relation becomes a function only when:

$$\forall x \in A, \exists ! y \in B$$

Meaning: For every input (x), there exists exactly one output (y).

6. Types of Functions

Functions are classified based on how inputs map to outputs.

Main types:

1. One-One Function (Injective)
2. Many-One Function
3. Onto Function (Surjective)
4. Into Function

Classification Overview

Function Type	Main Idea
One-One	Different inputs $\rightarrow$ different outputs
Many-One	Different inputs $\rightarrow$ same output
Onto	Every codomain element is used
Into	Some codomain elements are unused

1. One-One Function (Injective Function)

Definition

A function is called **One-One** if: Different inputs produce different outputs.
No two elements of domain have the same image.

Mathematical Condition

$$f(a) = f(b) \Rightarrow a = b$$

Example

$$f(x) = 2x$$

Domain:

{1,2,3}

Outputs:

x	(f(x))
1	2
2	4
3	6

All outputs are different.

Therefore: $f(x) = 2x$ is One-One.

Mapping Diagram

$$1 \rightarrow 2$$

$$2 \rightarrow 4$$

$$3 \rightarrow 6$$

No repeated output.

Not One-One Example

$$f(x) = x^2$$

For domain: {-2,-1,1,2}

Outputs:

x	(f(x))
-2	4
2	4

Different inputs give same output.

So NOT one-one.

Important Property

Property	Result
One-One + Onto	Bijjective
Inverse exists	Only for bijection

2. Many-One Function

Definition: A function is called **Many-One** if: Two or more inputs have the same output.

Example

$$f(x) = x^2$$

Domain: {-2,-1,1,2}

Outputs:

x	(f(x))
-2	4

2	4
-1	1
1	1

Different inputs map to same output.
Hence Many-One function.

Mapping Diagram

$-2 \rightarrow 4$

$2 \rightarrow 4$

$-1 \rightarrow 1$

$1 \rightarrow 1$

Important Note

Many-One functions are still valid functions because:

- each input has only one output.

3. Onto Function (Surjective Function)

Definition

A function is called **Onto** if: Every element of codomain has at least one pre-image.

Meaning: Every codomain element is actually used.

Mathematical Condition

$$\forall y \in B, \exists x \in A \text{ Such that } f(x) = y$$

Example

$$f(x) = x + 1$$

Domain: $A = \{1, 2, 3\}$

Codomain: $B = \{2, 3, 4\}$

Outputs:

x	(f(x))
1	2
2	3
3	4

Every codomain element is used.

Therefore Onto.

Visual Understanding

Codomain = $\{2, 3, 4\}$

Range = $\{2, 3, 4\}$

Range = Codomain

So Onto.

Non-Onto Example

Domain: $\{1, 2, 3\}$

Codomain: $\{2, 3, 4, 5\}$

Function: $f(x) = x + 1$

Outputs: $\{2, 3, 4\}$

5 is unused.

Therefore NOT onto.

Important Rule

Condition	Result
Range = Codomain	Onto
Range $\neq$ Codomain	Not Onto

4. Into Function

Definition: A function is called **Into** if: At least one element of codomain is not mapped by any domain element.

Example

Domain: $A = \{1, 2, 3\}$

Codomain: $B = \{2, 3, 4, 5\}$

Function: $f(x) = x + 1$

Outputs: $\{2, 3, 4\}$

Element 5 is unused.

Hence Into function.

Visual Understanding

Codomain = $\{2, 3, 4, 5\}$

Range = $\{2, 3, 4\}$

Since one codomain element is left unused $\rightarrow$ Into.

Comparison Table

Property	One-One	Many-One	Onto	Into
Different inputs give different outputs	✓	✗	Depends	Depends
Same output allowed	✗	✓	Possible	Possible
Every codomain element used	Depends	Depends	✓	✗
Range = Codomain	Not necessary	Not necessary	✓	✗

Combined Types

Combination	Meaning
One-One Onto	Bijjective Function
Many-One Onto	Multiple inputs share outputs but all codomain used
One-One Into	Different outputs but some codomain unused
Many-One Into	Shared outputs and unused codomain

Important GTU Exam Questions

Question Type	Example
Define One-One function	Theory
Difference between Onto and Into	Theory
Check function type	Numerical
Give example of Many-One	Conceptual
Determine whether function is onto	Problem solving
Draw mapping diagrams	Diagram-based

Fast Identification Tricks

Observation	Function Type
Repeated outputs	Many-One
All outputs different	One-One
Codomain fully covered	Onto
Some codomain unused	Into

Complete Concept Flow

Functions



One-One / Many-One



Onto / Into



Bijjective Function

7. Injective Function (One-One Function)

Definition: A function is called **Injective** if: Different inputs always produce different outputs. No two elements in the domain can map to the same element in the codomain.

Mathematical Definition

$$f(a) = f(b) \Rightarrow a = b$$

Equivalent meaning: If outputs are equal, then inputs must also be equal.

Simple Understanding

Different Input → Different Output

Example 1 : $f(x)=2x$

Domain: {1,2,3,4}

Outputs:

x	(f(x))
1	2
2	4
3	6
4	8

All outputs are different.
Therefore Injective.

Mapping Diagram

$1 \rightarrow 2$

$2 \rightarrow 4$

$3 \rightarrow 6$

$4 \rightarrow 8$

No repeated outputs.

Non-Injective Example

$$f(x) = x^2$$

Domain: {-2,-1,1,2}

Outputs:

x	(f(x))
-2	4
2	4

Two different inputs produce same output.

Therefore NOT injective.

Important Property

An injective function never has repeated outputs.

Graph Understanding

For real-valued functions: If a horizontal line cuts graph only once $\rightarrow$ Injective.

Important Notes

Point	Meaning
Injective = One-One	Same concept
Repeated output not allowed	Main condition
Inverse possible only if injective	Important theorem

8. Surjective Function (Onto Function)

Definition: A function is called **Surjective** if: Every element of codomain has at least one pre-image.

Meaning every codomain value is actually produced.

Mathematical Definition

$$\forall y \in B, \exists x \in A \text{ Such that } f(x) = y$$

Simple Understanding

Every codomain element is used.

Example

$$f(x) = x + 1$$

Domain: $A = \{1, 2, 3\}$

Codomain: $B = \{2, 3, 4\}$

Outputs:

x	(f(x))
1	2
2	3
3	4

All codomain elements are used.

Therefore Surjective.

Visual Representation

Codomain = $\{2, 3, 4\}$

Range = $\{2, 3, 4\}$

Range = Codomain

Hence Onto.

Non-Surjective Example

Domain: {1,2,3}

Codomain: {2,3,4,5}

Function: $f(x)=x+1$

Outputs: {2,3,4}

5 is unused.

Therefore NOT surjective.

Important Rule

Onto Condition

Range=Codomain

Important Notes

Point	Meaning
Surjective = Onto	Same concept
Every codomain element must appear	Main rule
Unused codomain not allowed	Important condition

9. Bijective Function

Definition

A function is called **Bijective** if it is:

1. Injective (One-One)
2. Surjective (Onto)

at the same time.

Mathematical Idea

Bijective= Injective + Surjective

Simple Understanding

Different inputs → Different outputs

AND

Every codomain element is used

Example

$f(x)=x+1$

Domain: $A=\{1,2,3\}$

Codomain: $B=\{2,3,4\}$

Outputs:

x	(f(x))
1	2
2	3
3	4

Properties:

Condition	Result
Different outputs	✓
All codomain used	✓

Therefore Bijective.

Mapping Diagram

$1 \rightarrow 2$

$2 \rightarrow 3$

$3 \rightarrow 4$

Perfect one-to-one mapping.

Non-Bijective Cases

Function Type	Why Not Bijective
Many-One Onto	Not injective
One-One Into	Not surjective
Many-One Into	Neither property

Most Important Property of Bijective Functions

A function has an inverse only if it is bijective.

Inverse Relation

$$f^{-1} \text{ exists} \iff f \text{ is Bijective}$$

Comparison Table

Property	Injective	Surjective	Bijective
Different outputs for different inputs	✓	Optional	✓
Every codomain element used	Optional	✓	✓
Range = Codomain	Not necessary	✓	✓
Repeated outputs allowed	✗	Possible	✗
Inverse exists	Partial condition	Partial condition	✓

Visual Summary

Injective: No repeated outputs

Surjective: No unused codomain elements

Bijective: No repeated outputs + No unused codomain elements

GTU Important Questions

Question Type	Example
Define injective function	Theory
Difference between onto and one-one	Theory
Determine bijective function	Numerical
Check whether inverse exists	Conceptual
Give example of surjective function	Example-based
Prove function is injective	Problem solving

Fast Identification Tricks

Observation	Type
Repeated outputs present	Not injective
Codomain fully used	Surjective
Both conditions satisfied	Bijective
Unused codomain exists	Into

10. Identity Function

Definition: An **Identity Function** is a function in which: Every element maps to itself.

Mathematical Representation

$$f(x)=x$$

Simple Meaning: Input and output are exactly same.

Example

Domain: $A=\{1,2,3,4\}$

Function: $f(x)=x$

Outputs:

x	(f(x))
1	1
2	2
3	3
4	4

Every value maps to itself.
Therefore it is Identity Function.

Mapping Diagram

$1 \rightarrow 1$

$2 \rightarrow 2$

$3 \rightarrow 3$

$4 \rightarrow 4$

Important Properties

Property	Result
One-One	✓
Onto	✓
Bijjective	✓
Inverse exists	✓
Inverse of identity	Identity itself

Identity Function Property

For any function (f):

$$f \circ I = I \circ f = f$$

Where $(I(x)=x)$.

Real-Life Analogy

Original data $\rightarrow$ unchanged output
No modification occurs.

11. Constant Function

Definition: A **Constant Function** is a function where: Every input gives the same output.

Mathematical Representation

$$f(x)=c$$

Where (c) is a constant.

Example

$$f(x)=5$$

Domain: {1,2,3,4}

Outputs:

x	(f(x))
1	5
2	5
3	5
4	5

All outputs are same.

Hence Constant Function.

Mapping Diagram

1 → 5

2 → 5

3 → 5

4 → 5

Important Properties

Property	Result
Many-One	✓
One-One	✗
Onto	Usually No
Range contains one element	✓

Key Observation

All outputs are repeated.

Therefore constant function is never injective (except single-element domain).

Identity vs Constant Function

Identity Function	Constant Function
Output changes with input	Same output always
(f(x)=x)	(f(x)=c)
Bijjective	Usually many-one
Different outputs	Same outputs

12. Composition of Functions

Definition: Composition means applying one function after another.

Mathematical Formula

$$(f \circ g)(x) = f(g(x))$$

Read as: "f composed with g"

Process

x

↓

g(x)

↓

f(g(x))

First apply (g), then apply (f).

Example 1

Given:

$$f(x)=2x+1$$

$$g(x) = x^2$$

Find:

$$(f \circ g)(x)$$

Step 1: Apply (g(x))

$$g(x) = x^2$$

Step 2: Put into (f)

$$\begin{aligned} f(g(x)) &= f(x^2) \\ &= 2x^2 + 1 \end{aligned}$$

Final Answer

$$(f \circ g)(x) = 2x^2 + 1$$

Example 2

Find:

$$(g \circ f)(x)$$

Step 1

$$f(x)=2x+1$$

Step 2

$$\begin{aligned} g(f(x)) &= (2x+1)^2 \\ &= 4x^2 + 4x + 1 \end{aligned}$$

Final Answer

$$(g \circ f)(x) = 4x^2 + 4x + 1$$

Important Observation

$$(f \circ g)(x) \neq (g \circ f)(x)$$

Usually composition is NOT commutative.

Composition Table

Step	Action
1	Apply inner function
2	Substitute into outer function
3	Simplify

Real-Life Analogy

Photo

↓

Filter App

↓

Compression App

Multiple operations performed sequentially.

13. Associative Property of Functions

Definition : Function composition follows associative law.

Formula

$$f \circ (g \circ h) = (f \circ g) \circ h$$

Meaning: Grouping does not matter. But order still matters.

Understanding

First grouping: $f \circ (g \circ h)$

Second grouping: $(f \circ g) \circ h$

Both give same result.

Example

Given:

$$f(x) = x + 1$$

$$g(x) = 2x$$

Complete Summary Table

Topic	Main Idea
Identity Function	$(f(x) = x)$
Constant Function	Same output for all inputs
Composition	Apply one function after another
Associative Property	Grouping does not matter

Important GTU Questions

Question Type	Example
Define identity function	Theory
Difference between identity and constant	Theory
Find composition	Numerical
Verify associative property	Problem solving
Show composition is not commutative	Proof
Evaluate $(f \circ g)$	Direct question

Most Important Concepts

Identity Function

$$I(x) = x$$

Constant Function

$$f(x) = c$$

Composition

$$(f \circ g)(x) = f(g(x))$$

Associative Property

$$f \circ (g \circ h) = (f \circ g) \circ h$$

14. Inverse Function

Definition

An **inverse function** reverses the action of a function.

If: $f(a)=b$

then: $f^{-1}(b) = a$

Simple Meaning

Function: Input $\rightarrow$ Output

Inverse Function: Output $\rightarrow$ Input

Important Condition

A function has an inverse only if it is:

1. One-One (Injective)
2. Onto (Surjective)

That means function must be bijective.

Inverse Function Condition

$$f^{-1} \text{ Exists} \iff f \text{ Is Bijective}$$

Example

Given:

$$f(x)=2x+3$$

If: $x=1$

then: $f(1)=5$

Inverse reverses it:

$$f^{-1}(5) = 1$$

Visual Understanding

$f(x): 1 \rightarrow 5$

$$f^{-1}(x): 5 \rightarrow 1$$

Important Properties

Property	Result
Exists only for bijection	✓
Reverses mapping	✓
Domain and range interchange	✓

Domain and Range Swap

Function	Inverse Function
Domain	Range
Range	Domain

15. Finding Inverse Function

Replace x and y Method

This is the standard GTU method.

Steps to Find Inverse

Step	Action
1	Write $(y=f(x))$
2	Replace $(f(x))$ with (y)

3	Swap (x) and (y)
4	Solve for (y)
5	Replace (y) by $(f^{-1})(x)$

Example 1

Find inverse of: $f(x)=2x+3$

Step 1

Write: $y=2x+3$

Step 2

Swap (x) and (y)

$$x=2y+3$$

Step 3

Solve for (y)

$$x-3=2y$$

$$y = \frac{x-3}{2}$$

Final Answer

$$f^{-1}(x) = \frac{x-3}{2}$$

Example 2

Find inverse of:

$$f(x) = \frac{x+1}{2}$$

Step 1

$$y = \frac{x+1}{2}$$

Step 2

Swap variables

$$x = \frac{y+1}{2}$$

Step 3

Solve for (y)

$$2x=y+1$$

$$y=2x-1$$

Final Answer

$$f^{-1}(x) = 2x - 1$$

Important Note

If function is not one-one, inverse may not exist.

Example of Non-Invertible Function

$$f(x) = x^2$$

Because:

$$f(2)=4$$

and

$$f(-2)=4$$

Same output for different inputs.

So inverse is not uniquely defined.

16. Verification of Inverse Function

After finding inverse, verify it.

Verification Formula

$$f(f^{-1}(x)) = x$$

and also

$$f^{-1}(f(x)) = x$$

Example Verification

Given: $f(x)=2x+3$

Inverse:

$$f^{-1}(x) = \frac{x-3}{2}$$

Step 1

Find:

$$f(f^{-1}(x))$$

Substitute inverse:

$$f\left(\frac{x-3}{2}\right)$$

Step 2

Apply function:

$$= 2\left(\frac{x-3}{2}\right) + 3$$

$$= x - 3 + 3$$

$$= x$$

Verified.

Second Verification

$$f^{-1}(f(x))$$

Substitute function:

$$f^{-1}(2x+3)$$

Apply inverse:

$$= \frac{(2x+3)-3}{2}$$

$$= \frac{2x}{2}$$

=x

Verified again.

Important Meaning

Function + Inverse
returns original value

Important Properties

Property	Result
Inverse reverses function	✓
Composition gives identity	✓
Both verifications should give x	✓

Identity Relation

$$f \circ f^{-1} = I$$

17. Applications of Functions

Functions are heavily used in Computer Science, Mathematics, AI, and Engineering.

A. Database Mapping

Input	Output
Student ID	Student Record
Employee ID	Employee Data

Each ID maps to information.

B. Password Systems

Input Password

↓

Hash Function

↓

Encrypted Output

Functions convert original data into secure form.

C. Coding Theory

Functions are used in:

- Error detection
- Error correction
- Encoding systems

D. Artificial Intelligence

Functions are used in:

Area	Usage
Neural Networks	Activation functions
Machine Learning	Prediction functions
Optimization	Cost functions

E. Operating Systems

Functions help in:

- Address mapping
- Resource allocation
- Scheduling systems

F. Graph Theory

Functions map:

- Nodes
- Edges
- Relationships

G. Digital Electronics

Logic gates behave like functions.

Input	Output
Binary values	Logic result

H. Compiler Design

Functions are used in:

- Symbol tables
- Parsing
- Translation

I. Cryptography

Encryption and decryption use mathematical functions.

Real-Life Examples

Function	Application
Temperature conversion	Celsius → Fahrenheit
ATM system	PIN → Account access
GPS	Coordinates → Location
Search engine	Query → Results

Complete Summary Table

Topic	Main Idea
Inverse Function	Reverses original function
Finding Inverse	Swap x and y
Verification	Composition gives x
Applications	Mapping, coding, AI, systems

Important GTU Questions

Question Type	Example
Define inverse function	Theory
Find inverse	Numerical
Verify inverse	Problem solving
Conditions for inverse	Conceptual
Applications of functions	Short notes
Why inverse may not exist	Reasoning

Most Important Formulas

Inverse Existence

$$f^{-1} \text{ Exists} \iff f \text{ Is Bijective}$$

Verification

$$f(f^{-1}(x)) = x$$

Composition Identity

$$f \circ f^{-1} = I$$

Video Based Learning:

https://youtube.com/playlist?list=PLEHGYFbPuuMHwxnYr_c8nu4Zm1rcXt1bq&si=Ef18a-Ei4JzA1h-b/

Algebraic Structures — Basics

Algebraic structures are mathematical systems formed using:

1. A set
2. One or more operations

These concepts are very important in:

- Cryptography
- Coding Theory
- Computer Science
- Automata Theory
- Data Structures

Video Based Learning:

https://youtube.com/playlist?list=PLEHGYFbPuuMFliZ3cxhMf9eL4bKpjqlFC&si=O0SGP_s_mxxBYET/

1. Binary Operation

Definition: A **Binary Operation** is an operation performed on:

Two elements of a set to produce another element of the same set.

Mathematical Representation

If (S) is a set, then binary operation is:

$$*: S \times S \rightarrow S$$

Meaning:

- Take two elements from set (S)
- Apply operation
- Result must also belong to (S)

Simple Understanding

$$a * b = c$$

where:

$$a \in S$$

$$b \in S$$

$$c \in S$$

Most Important Condition

Closure Property: Result must stay inside the set.

Closure Formula

$$a, b \in S \Rightarrow a * b \in S$$

Example 1 — Addition on Integers

Set:

$$Z = \dots, -2, -1, 0, 1, 2, \dots$$

Operation: +

Take two integers:

$$2 + 3 = 5$$

5 is also an integer.

Therefore addition is a binary operation on integers.

Example 2 — Multiplication on Natural Numbers

Set:

$$N = 1, 2, 3, \dots$$

Operation: $\times$

Take:

$$2 \times 4 = 8$$

8 belongs to natural numbers.

Hence binary operation.

Example 3 — Not a Binary Operation

Set:

$$N = 1, 2, 3, \dots$$

Operation:

$$\text{Take: } 2 - 5 = -3$$

$$(-3 \notin N)$$

Closure fails.

Therefore subtraction is NOT a binary operation on natural numbers.

Important Examples Table

Set	Operation	Binary Operation?	Reason
Integers	Addition	✓	Closed
Integers	Multiplication	✓	Closed
Natural Numbers	Subtraction	✗	Negative result possible
Real Numbers	Division	✗	Division by zero issue

Binary Operation Properties

A binary operation may satisfy:

Property	Meaning
Closure	Result inside set
Associative	Grouping unchanged
Commutative	Order unchanged
Identity	Neutral element exists
Inverse	Opposite element exists

These properties help form advanced algebraic structures.

Common Symbols Used

Symbol	Operation
+	Addition
$\times$	Multiplication
*	General operation
$\circ$	Composition

Real-Life Analogy

Input 1 + Input 2

↓

Operation

↓

Output in same system

Example:

- Two bank balances added
- Two binary bits processed
- Two matrices multiplied

2. Algebraic Structure

Definition: An **Algebraic Structure** is: A non-empty set together with one or more binary operations.

Basic Representation

$(S, *)$

Where:

Symbol	Meaning
(S)	Set
$(*)$	Binary operation

Simple Meaning

Set +

Operation =

Algebraic Structure

Example 1

$(\mathbb{Z}, +)$

Meaning:

- Set = Integers
- Operation = Addition

This forms an algebraic structure.

Example 2

$(\mathbb{N}, \times)$

Set = Natural numbers

- Operation = Multiplication

Also an algebraic structure.

Why Algebraic Structures Matter

They help study:

Area	Usage
Cryptography	Encryption systems
Coding Theory	Error correction
Computer Science	Automata, FSM
AI	Mathematical models
Graph Theory	Structural analysis

Types of Algebraic Structures

As more properties are added, stronger structures form.

Hierarchy

Set + Binary Operation



Semigroup



Monoid



Group



Abelian Group

Basic Structure Requirements

Structure	Required Properties
Binary Operation	Closure
Semigroup	Closure + Associative
Monoid	Semigroup + Identity
Group	Monoid + Inverse
Abelian Group	Group + Commutative

Important Properties

Associative Property

$$(ab)c = a(bc)$$

Commutative Property

$$ab = ba$$

Identity Property

$$ae = ea = a$$

Inverse Property

$$a * a^{-1} = e$$

Example Analysis

Consider:

$$(Z, +)$$

Check properties:

Property	Result
Closure	✓
Associative	✓

Identity	0
Inverse	-a exists
Commutative	✓

Thus it becomes an Abelian Group.

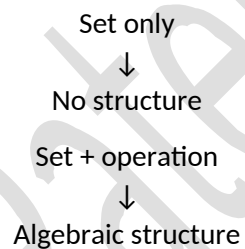
Important Difference

Binary Operation	Algebraic Structure
Only operation	Set + operation
Focus on closure	Focus on properties
Basic concept	Complete mathematical system

GTU Important Questions

Question Type	Example
Define binary operation	Theory
Explain closure property	Conceptual
Give examples of binary operation	Examples
Define algebraic structure	Theory
Check whether operation is binary	Numerical
Difference between set and algebraic structure	Theory

Fast Understanding Trick



Complete Summary Table

Topic	Main Idea
Binary Operation	Operation on two set elements
Closure	Result remains in set
Algebraic Structure	Set + operation
Properties	Associative, identity, inverse

Most Important Concepts

Binary Operation *

$$\times S \rightarrow S$$

Closure Property

$$a, b \in S \Rightarrow a * b \in S$$

Algebraic Structure (S, *)

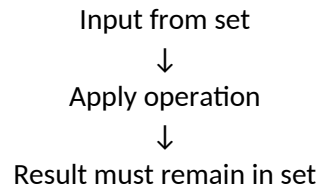
3. Closure Property

Definition: A set is said to satisfy the **Closure Property** under an operation if: Performing the operation on any two elements of the set gives a result that also belongs to the same set.

Mathematical Representation

$$a, b \in S \Rightarrow a * b \in S$$

Simple Meaning



Example 1 — Integers Under Addition

Set:

$$\mathbb{Z} = \dots, -2, -1, 0, 1, 2, \dots$$

Operation: +

Take: $3+5=8$

8 is also an integer.

Therefore closure property holds.

Example 2 — Natural Numbers Under Multiplication

$$2 \times 4 = 8$$

8 belongs to natural numbers.

Closure satisfied.

Example 3 — Natural Numbers Under Subtraction

$$2-5=-3$$

$$(-3 \notin \mathbb{N})$$

Closure fails.

Hence subtraction is not closed on natural numbers.

Closure Table

Set	Operation	Closed?
Integers	Addition	✓
Integers	Multiplication	✓
Natural Numbers	Subtraction	✗
Real Numbers	Addition	✓

Important Note

Closure is the most basic requirement for any algebraic structure.

Without closure:

- semigroup cannot exist
- monoid cannot exist
- group cannot exist

Real-Life Analogy

Currency conversion inside same system

↓

Result still valid currency

4. Associative Property

Definition: An operation is associative if: Changing grouping does not change the result.

Mathematical Formula

$$(ab)c=a(bc)$$

Simple Meaning

Grouping changes BUT Answer remains same

Example 1 — Addition

$$(2+3)+4$$

$$=5+4=9$$

and

$$2+(3+4)$$

$$=2+7=9$$

Both equal.

Therefore addition is associative.

Example 2 — Multiplication

$$(2 \times 3) \times 4 \\ = 6 \times 4 = 24$$

and

$$2 \times (3 \times 4) \\ = 2 \times 12 = 24$$

Associative property holds.

Example 3 — Subtraction

$$(5-3)-2$$

$$=2-2=0$$

and

$$5-(3-2)$$

$$=5-1=4$$

Results differ.

Therefore subtraction is NOT associative.

Associative Property Table

Operation	Associative?
Addition	✓
Multiplication	✓
Subtraction	✗
Division	✗

Important Difference

Associative	Commutative
Grouping changes	Order changes
$((a*b)*c)$	$(ab=ba)$

Students often confuse them.

Real-Life Analogy

Packing items in different groups

↓

Total weight unchanged

5. Identity Element

Definition: An **Identity Element** is an element that: Does not change another element after operation.

Mathematical Formula

$$ae=ea=a$$

Where:

- (e) = identity element

Simple Meaning

Apply operation with identity

↓

Original value remains unchanged

Example 1 — Addition

Operation: +

Identity element: 0

Because: $a+0=0+a=a$

Example 2 — Multiplication

Identity element: 1

Because:

$$a \times 1 = 1 \times a = a$$

Example Table

Operation	Identity Element
Addition	0
Multiplication	1

Important Notes

Point	Meaning
Identity is unique	Only one identity exists
Every monoid/group needs identity	Important condition
Identity keeps value unchanged	Main idea

Example Verification

For integers under addition:

Take: $a=5$

Check: $5+0=5$

Identity verified.

Real-Life Analogy

Adding zero money

↓

Balance unchanged

6. Inverse Element

Definition: An **Inverse Element** is an element that: Produces identity when operated with another element.

Mathematical Formula

$$a * a^{-1} = e$$

Where:

- $(a^{-1}) = \text{inverse of } (a)$
- $(e) = \text{identity}$

Simple Meaning

Element

+

Opposite element

↓

Identity obtained

Example 1 — Addition

Set:

$$(Z, +)$$

Identity: 0

Take: $a=5$

Inverse: -5

Because: $5+(-5)=0$

Example 2 — Multiplication

Take: $a=2$

Inverse:

$$\frac{1}{2}$$

Because:

$$2 \times \frac{1}{2} = 1$$

Inverse Table

Operation	Identity	Inverse of (a)
Addition	0	$(-a)$
Multiplication	1	$(\frac{1}{a})$

Important Notes

Point	Meaning
Inverse depends on identity	Important
Every group element must have inverse	Group condition
Inverse “undoes” operation	Main idea

Example of No Inverse

Set:

$$N = 1,2,3,...$$

Operation: +

Take: 5

Inverse should be: -5

But:

$$-5 \notin N$$

Therefore inverse does not exist in natural numbers under addition.

Identity vs Inverse

Identity Element	Inverse Element
Neutral element	Opposite element
Keeps value same	Produces identity
Example: 0	Example: $-a$

Complete Summary Table

Property	Main Idea
Closure	Result remains in set
Associative	Grouping unchanged
Identity	Neutral element
Inverse	Opposite element

Structure Formation Flow

Closure

↓

Associative

↓

Identity

↓

Inverse

↓

Group Formation

GTU Important Questions

Question Type	Example
Define closure property	Theory
Check associative property	Numerical
Find identity element	Direct question
Find inverse element	Numerical
Difference between identity and inverse	Theory
Give examples	Conceptual

Most Important Formulas

Closure

$$a, b \in S \Rightarrow a * b \in S$$

Associative

$$(ab)c = a(bc)$$

Identity

$$ae = ea = a$$

Inverse

$$a * a^{-1} = e$$

7. Commutative Property

Definition: An operation is called **Commutative** if: Changing the order of elements does not change the result.

Mathematical Formula

$$ab=ba$$

Simple Meaning

Order changes BUT Answer remains same

Example 1 — Addition

$$2+5=7$$

$$5+2=7$$

Both answers are same.

Therefore addition is commutative.

Example 2 — Multiplication

$$3 \times 4 = 12$$

$$4 \times 3 = 12$$

Multiplication is commutative.

Example 3 — Subtraction

$$5-2=3$$

$$2-5=-3$$

Results differ.

Therefore subtraction is NOT commutative.

Example 4 — Division

$$8 \div 2 = 4$$

$$2 \div 8 = \frac{1}{4}$$

Different answers.

So division is NOT commutative.

Commutative Property Table

Operation	Commutative?
Addition	✓
Multiplication	✓
Subtraction	✗
Division	✗

Important Difference

Associative	Commutative
Grouping changes	Order changes
$((a*b)*c)$	$(ab=ba)$

Real-Life Analogy

2 apples + 3 apples = 3 apples + 2 apples

Total remains same.

8. Semigroup

Definition: A **Semigroup** is an algebraic structure in which:

1. Closure property holds
2. Associative property holds

Mathematical Representation

$(S, *)$

such that:

1. Closure
2. Associativity

Semigroup Conditions

Property	Required?
Closure	✓
Associative	✓
Identity	✗
Inverse	✗

Core Formula

Associative Condition

$$(ab)c = a(bc)$$

Example 1 — Natural Numbers Under Addition

$$(N, +)$$

Check:

Property	Result
Closure	✓
Associative	✓

Therefore semigroup.

Example 2 — Positive Integers Under Multiplication

$$(Z^+, \times)$$

Closure and associativity both hold.

Hence semigroup.

Example of NOT Semigroup

$$(N, -)$$

Subtraction is not associative.

So not semigroup.

Important Notes

Point	Meaning
Simplest algebraic structure	✓
Identity not necessary	✓
Inverse not necessary	✓

Structure Flow

Set +

Binary Operation +
Closure +
Associative
=
Semigroup

9. Monoid

Definition: A **Monoid** is a semigroup with an identity element.

Conditions of Monoid

Property	Required?
Closure	✓
Associative	✓
Identity	✓
Inverse	✗

Mathematical Structure

$(S, *)$

must satisfy:

1. Closure
2. Associativity
3. Identity element

Identity Formula

$ae=ea=a$

Example 1 — Integers Under Addition

$(\mathbb{Z}, +)$

Check properties:

Property	Result
Closure	✓
Associative	✓
Identity (0)	✓

Therefore monoid.

Example 2 — Natural Numbers Under Multiplication

$(\mathbb{N}, \times)$

Identity: 1

All conditions satisfied.

Hence monoid.

Example of NOT Monoid

$(\mathbb{N}, -)$

No associativity.

Hence not monoid.

Important Notes

Point	Meaning
Every monoid is semigroup	✓

Identity compulsory	✓
Inverse not compulsory	✓

Structure Flow

Semigroup + Identity = Monoid

10. Group

Definition : A **Group** is a monoid in which every element has an inverse.

Conditions of Group

Property	Required?
Closure	✓
Associative	✓
Identity	✓
Inverse	✓

Mathematical Representation

$(G, *)$

Inverse Formula

$$a * a^{-1} = e$$

Example 1 — Integers Under Addition

$(\mathbb{Z}, +)$

Check:

Property	Result
Closure	✓
Associative	✓
Identity (0)	✓
Inverse exists	✓

Inverse of: $5 = -5$

Therefore group.

Example 2 — Nonzero Real Numbers Under Multiplication

$(\mathbb{R} - 0, \times)$

Identity: 1

Inverse of:

$$a = \frac{1}{a}$$

All conditions satisfied.

Hence group.

Example of NOT Group

$(\mathbb{N}, +)$

Inverse of 5 is: -5

But:

$$-5 \notin \mathbb{N}$$

Inverse property fails.

Therefore not group.

Important Notes

Point	Meaning
Every group is monoid	✓
Inverse compulsory	✓
Identity compulsory	✓

Abelian Group

If group is also commutative: $ab=ba$
then it becomes an **Abelian Group**.

Example

$$(\mathbb{Z}, +)$$

Addition is commutative.
Hence Abelian Group.

Hierarchy of Structures

Semigroup



Monoid



Group



Abelian Group

Comparison Table

Property	Semigroup	Monoid	Group
Closure	✓	✓	✓
Associative	✓	✓	✓
Identity	✗	✓	✓
Inverse	✗	✗	✓

Complete Structure Example Integers Under Addition

$$(\mathbb{Z}, +)$$

Property	Result
Closure	✓
Associative	✓
Identity	0
Inverse	-a
Commutative	✓

Therefore:

- Semigroup ✓
- Monoid ✓
- Group ✓
- Abelian Group ✓

Real-Life Analogy

Structure	Analogy
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Semigroup	Basic machine
Monoid	Machine with reset button
Group	Machine with undo feature

GTU Important Questions

Question Type	Example
Define semigroup	Theory
Difference between monoid and group	Theory
Check whether structure is group	Numerical
Find identity and inverse	Practical
Define Abelian group	Theory
Verify associative property	Problem solving

Most Important Formulas

Commutative Property

$$ab=ba$$

Associative Property

$$(ab)c=a(bc)$$

Identity Property

$$ae=ea=a$$

Inverse Property

$$a * a^{-1} = e$$

11. Abelian Group

Definition: An **Abelian Group** is a group in which: Commutative property also holds.

Conditions of Abelian Group

A structure must satisfy:

1. Closure
2. Associative property
3. Identity element
4. Inverse element
5. Commutative property

Commutative Formula

$$ab=ba$$

Structure Representation

$$(G, *)$$

Example 1 — Integers Under Addition

$$(Z, +)$$

Check:

Property	Result
Closure	✓
Associative	✓

Identity	0
Inverse	-a
Commutative	✓

Therefore Abelian Group.

Example 2 — Real Numbers Under Addition

$$(R, +)$$

Addition is commutative.

Hence Abelian Group.

Example of Non-Abelian Group

Matrix multiplication.

Because:

$$AB \neq BA$$

Generally not commutative.

Therefore non-Abelian.

Important Notes

Point	Meaning
Every Abelian group is a group	✓
Every group is not Abelian	✓
Commutativity is key difference	✓

Real-Life Analogy

Changing order does not affect result

12. Finite Group

Definition : A **Finite Group** is a group containing: Finite number of elements.

Mathematical Idea

If: $|G|=n$

where (n) is finite, then (G) is finite group.

Example 1

$$G=\{0,1,2,3\}$$

under addition modulo 4.

Number of elements:

$$|G|=4$$

Hence finite group.

Example 2

$$Z_5 = 0,1,2,3,4$$

Finite because only 5 elements exist.

Important Notes

Point	Meaning
Number of elements countable	✓
Group size limited	✓
Used heavily in coding theory	✓

Real-Life Analogy

Fixed number of objects
inside system

13. Infinite Group

Definition: An **Infinite Group** is a group containing: Infinite number of elements.

Example 1 — Integers Under Addition

$$(\mathbb{Z}, +)$$

Elements:

..., -3, -2, -1, 0, 1, 2, 3, ...

Infinite elements exist.

Therefore infinite group.

Example 2 — Real Numbers Under Addition

$$(\mathbb{R}, +)$$

Infinite elements.

Hence infinite group.

Important Difference

Finite Group	Infinite Group
Limited elements	Unlimited elements
Count finite	Count infinite

Examples Table

Group	Type
$(\mathbb{Z}_4)$	Finite
$((\mathbb{Z}, +))$	Infinite
$((\mathbb{R}, +))$	Infinite

14. Order of Group

Definition: The **Order of a Group** means: Number of elements present in the group.

Mathematical Representation

$$|G| = n$$

Where:

- (G) = group
- (n) = number of elements

Example 1

$$G = \{0, 1, 2, 3\}$$

Then:

$$|G| = 4$$

Example 2

$$\mathbb{Z}_5 = 0, 1, 2, 3, 4$$

Order:

$$|G| = 5$$

Infinite Order

For integers:

$$(\mathbb{Z}, +)$$

Order is infinite.

Order of Element

Sometimes “order” also means: Number of times an element must operate with itself to produce identity.

Example

In:

$$(\mathbb{Z}_4, +)$$

Element: 2

Check:

$$2 + 2 = 4 \equiv 0 \pmod{4}$$

Identity obtained after 2 operations.

Therefore order of element 2 is: 2

Important Notes

Point	Meaning
Order of group = total elements	✓
Order of element = repeated operation count	✓
Important in Lagrange theorem	✓

15. Cyclic Group

Definition: A **Cyclic Group** is a group that can be generated using: A single element. That element is called generator.

Mathematical Representation

$$G = \langle a \rangle$$

Read as: “Group generated by (a)”

Core Idea

Repeated operation of one element produces all elements of group.

Example 1 — Integers Modulo 4

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$

Take generator: 1

Repeated addition modulo 4:

Step	Result
(1)	1
(1+1)	2
(1+1+1)	3
(1+1+1+1)	0

All elements generated.

Hence cyclic group.

Generator Representation

$$G = \langle a \rangle$$

Example 2 — Integers Under Addition

$$(\mathbb{Z}, +)$$

Generated by: 1

because:

..., -2, -1, 0, 1, 2, ...

can be formed using repeated addition/subtraction of 1.

Hence cyclic.

Important Notes

Point	Meaning
Generated by one element	✓
Generator creates entire group	✓
Every cyclic group is Abelian	✓

Cyclic Group Types

Type	Example
Finite cyclic	$(\mathbb{Z}_4)$
Infinite cyclic	$((\mathbb{Z}, +))$

Important Property

Every cyclic group is commutative.

Example of Generator

In:

$$\mathbb{Z}_5$$

Generator: 1

because repeated addition generates all elements.

Not Every Element Is Generator

Example:

In:

$$\mathbb{Z}_4$$

Element: 2

Repeated addition:

2, 0, 2, 0, ...

Cannot generate all elements.

So 2 is not generator.

Hierarchy Summary

Group

↓

Abelian Group

↓

Cyclic Group

Comparison Table

Structure	Main Property
Abelian Group	Commutative
Finite Group	Finite elements
Infinite Group	Infinite elements

Cyclic Group	Generated by one element
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Important GTU Questions

Question Type	Example
Define Abelian group	Theory
Difference between finite and infinite group	Theory
Find order of group	Numerical
Define cyclic group	Theory
Find generator	Problem solving
Prove cyclic group is Abelian	Conceptual

Most Important Formulas

Commutative Property

$$ab=ba$$

Order of Group

$$|G|=n$$

Cyclic Group

$$G = \langle a \rangle$$

16. Generator

Definition: A **Generator** is an element of a cyclic group that:
Produces all elements of the group using repeated operation.

Mathematical Representation

If element (a) generates group (G):

$$G = \langle a \rangle$$

Simple Meaning

One element

↓

Repeated operation

↓

Entire group generated

Example 1 — $\mathbb{Z}_4$

Group:

$$\mathbb{Z}_4 = \{0,1,2,3\}$$

Take element: 1

Repeated addition modulo 4:

Operation	Result
(1)	1
(1+1)	2
(1+1+1)	3
(1+1+1+1)	0

All elements obtained.

Therefore 1 is generator.

Example 2 — Another Generator

In:

$$\mathbb{Z}_5$$

Element 2:

Operation	Result
2	2
2+2	4
2+2+2	1
2+2+2+2	3
2+2+2+2+2	0

All elements generated.

So 2 is also generator.

Non-Generator Example

In:

$$\mathbb{Z}_4$$

Take: 2

Repeated addition:

Operation	Result
2	2
2+2	0
2+2+2	2

Only 0 and 2 obtained.

Therefore 2 is not generator.

Important Notes

Point	Meaning
Generator creates full group	✓
Cyclic group may have multiple generators	✓
Not every element is generator	✓

17. Subgroup

Definition: A **Subgroup** is: A smaller group inside a larger group.

If:

$$H \subseteq G$$

and (H) itself satisfies all group properties, then (H) is subgroup of (G).

Symbol:

$$H \leq G$$

Conditions for Subgroup

A subset must satisfy:

1. Closure
2. Associativity
3. Identity
4. Inverse

Simple Understanding

Large Group

↓

Smaller part

↓

Still behaves like group

Example 1

Group:

$$(\mathbb{Z}, +)$$

Subset:

$$2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$$

Check:

Property	Result
Closure	✓
Identity (0)	✓
Inverse exists	✓

Therefore subgroup.

Example 2

$\{0\}$

inside:

$$(\mathbb{Z}, +)$$

Also subgroup.

Important Notes

Point	Meaning
Every subgroup is group	✓
Subgroup must use same operation	✓
Identity of subgroup same as parent group	✓

Subgroup Test

A non-empty subset (H) is subgroup if:

$$a, b \in H \Rightarrow ab^{-1} \in H$$

18. Proper Subgroup

Definition: A **Proper Subgroup** is a subgroup that is: Not equal to the whole group.

Mathematical Representation

$H < G$

meaning:

$$H \subset G$$

and

$$H \neq G$$

Example

Group:

$$(\mathbb{Z}, +)$$

Subgroup:

$$2\mathbb{Z}$$

Since:

$$2\mathbb{Z} \neq \mathbb{Z}$$

it is proper subgroup.

Non-Example G

itself is subgroup of (G), but not proper subgroup.

Important Notes

Point	Meaning
Every proper subgroup is subgroup	✓
Whole group not proper	✓

Visual Understanding

Group G

└─ Smaller subgroup H

19. Trivial Subgroup

Definition: A **Trivial Subgroup** is the subgroup containing: Only the identity element.

Representation

{e}

Where (e) is identity.

Example

In:

$$(\mathbb{Z}, +)$$

Identity: 0

Trivial subgroup: {0}

Why It Is Group

Check:

Property	Result
Closure	✓
Identity	✓
Inverse	✓

Hence subgroup.

Important Notes

Point	Meaning
Every group has trivial subgroup	✓
Smallest subgroup possible	✓

Types of Basic Subgroups

Type	Example
Trivial subgroup	{0}
Proper subgroup	$(2\mathbb{Z})$
Whole subgroup	$(G) \text{ itself}$

20. Normal Subgroup

Definition : A subgroup (N) of group (G) is called **Normal Subgroup** if: Left cosets and right cosets are equal.

Symbol

$$N \trianglelefteq G$$

Core Condition

$$gN = Ng$$

for every:

$$g \in G$$

Simple Understanding

Changing side

does not change coset

Example 1 — Abelian Group

Every subgroup of an Abelian group is normal.

Example:

$$(\mathbb{Z}, +)$$

Subgroup:

$$2\mathbb{Z}$$

Normal subgroup.

Why?

Because addition is commutative.

Example 2

In:

$$(\mathbb{R}, +)$$

every subgroup is normal.

Important Notes

Point	Meaning
Normal subgroup needed for quotient groups	✓
All Abelian groups have normal subgroups	✓
Every normal subgroup is subgroup	✓

Difference Between Subgroup and Normal Subgroup

Subgroup	Normal Subgroup
Smaller group inside group	Special subgroup
Cosets may differ	Left and right cosets equal

Visual Understanding

Subgroup

↓

Special condition

↓

Normal subgroup

Hierarchy

Group

↓

Subgroup

↓

Normal Subgroup

Complete Summary Table

Topic	Main Idea
Generator	Creates whole cyclic group
Subgroup	Smaller group inside group
Proper Subgroup	Subgroup not equal to whole group
Trivial Subgroup	Contains only identity
Normal Subgroup	Left and right cosets equal

Important GTU Questions

Question Type	Example
Define subgroup	Theory
Define proper subgroup	Theory
What is trivial subgroup	Direct question
Explain normal subgroup	Theory
Find generator of cyclic group	Numerical
Check subgroup properties	Problem solving

Most Important Formulas

Generator

$$G = \langle a \rangle$$

Subgroup Test

$$a, b \in H \Rightarrow ab^{-1} \in H$$

Normal Subgroup

$$gN = Ng$$

21. Cosets

Definition: A Coset is formed by:

Multiplying every element of a subgroup by a fixed group element.

Cosets help divide a group into equal-sized parts.

Types of Cosets

Type	Formula
Left Coset	(aH)
Right Coset	(Ha)

Where:

- (H) = subgroup
- (a) = element of group

Left Coset Formula

$$aH = \{ah \in H\}$$

Right Coset Formula

$$Ha = \{ha \in H\}$$

Simple Understanding

Take subgroup

↓

Shift using group element

↓

New set formed = coset

Example

Group:

$$(\mathbb{Z}, +)$$

Subgroup:

$$H = 2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$$

Take:

$$1 + H$$

Then:

$$1 + H = \{\dots, -3, -1, 1, 3, 5, \dots\}$$

This is a coset.

Important Notes

Point	Meaning
Cosets partition group	✓
All cosets have same size	✓
Used in Lagrange theorem	✓

Normal Subgroup Relation

If subgroup is normal: $aH = Ha$

Visual Understanding

Group divided into equal blocks

↓

Each block = coset

22. Homomorphism

Definition: A **Homomorphism** is a mapping between two algebraic structures that: Preserves the operation.

Mathematical Formula

$$f(a * b) = f(a) \circ f(b)$$

Meaning

Operation before mapping : Operation after mapping

Structure Representation

$$f: G \rightarrow G'$$

Where:

- (G) and (G') are groups

Simple Understanding

Structure changes BUT operation behavior preserved

Example 1

Define:

$$f: \mathbb{Z} \rightarrow \mathbb{Z}_n$$

by:

$$f(x) = x \bmod n$$

Check:

$$f(a + b) = f(a) + f(b)$$

Homomorphism holds.

Example 2

$$f(x) = 2x$$

Check:

$$f(a+b) = 2(a+b)$$

$$= 2a + 2b$$

$$= f(a) + f(b)$$

Hence homomorphism.

Important Notes

Point	Meaning
Structure-preserving map	✓
Operation remains consistent	✓
Important in algebra and coding	✓

Types of Homomorphism

Type	Meaning
Monomorphism	One-One homomorphism
Epimorphism	Onto homomorphism
Isomorphism	One-One + Onto homomorphism

23. Kernel

Definition: The **Kernel** of a homomorphism is: Set of all elements mapped to identity element.

Mathematical Formula

$$\text{Ker}(f) = \{x \in G : f(x) = e'\}$$

Where:

- (e') = identity of target group

Simple Understanding

Elements becoming identity
after mapping

Example

Define:

$$f: \mathbb{Z} \rightarrow \mathbb{Z}_4$$

by:

$$f(x) = x \pmod{4}$$

Find kernel.

Identity in:

$$\mathbb{Z}_4$$

is: 0

Now find:

$$x \pmod{4} = 0$$

Solutions:

$$\dots, -8, -4, 0, 4, 8, \dots$$

Thus:

$$\text{Ker}(f) = 4\mathbb{Z}$$

Important Properties

Property	Result
Kernel is subgroup	✓
Kernel is normal subgroup	✓
Trivial kernel $\Rightarrow$ injective homomorphism	✓

Important Note

Kernel measures: How much information is lost during mapping.

Visual Understanding

Mapped to identity

↓

Kernel elements

24. Isomorphism

Definition

An **isomorphism** is a homomorphism that is:

1. One-One
2. Onto

Mathematical Idea

Isomorphism = Homomorphism + Bijective

Meaning

Two algebraic structures are essentially identical.

Only element names differ.

Structure Representation

$$G \cong H$$

Read as:

"G is isomorphic to H"

Simple Understanding

Different appearance BUT same structure

Example

$$(\mathbb{R}^{\{+\}}, \times)$$

and

$$(\mathbb{R}, +)$$

are isomorphic under logarithm mapping.

Because:

$$\log(ab) = \log a + \log b$$

important Properties

Property	Result
Structure preserved	✓
Bijjective mapping	✓
Operations correspond	✓

Isomorphism Formula

$$f(a * b) = f(a) \circ f(b)$$

plus:

- injective
- surjective

Difference Between Homomorphism and Isomorphism

Homomorphism	Isomorphism
Structure preserving	Structure preserving
Need not be bijective	Must be bijective
Partial similarity	Complete similarity

Important Notes

Point	Meaning
Isomorphic groups behave same	✓
Structure more important than symbols	✓
Important in abstract algebra	✓

Hierarchy

Homomorphism

↓

Injective + Surjective

↓

Isomorphism

Complete Summary Table

Topic	Main Idea
Cosets	Shifted subgroup
Homomorphism	Structure-preserving map
Kernel	Elements mapped to identity
Isomorphism	Bijjective homomorphism

Important GTU Questions

Question Type	Example
Define coset	Theory
Difference between left and right coset	Theory

Define homomorphism	Theory
Find kernel	Numerical
Define isomorphism	Theory
Difference between homomorphism and isomorphism	Conceptual

Most Important Formulas

Left Coset

$$aH = \{ah \in H\}$$

Homomorphism

$$f(a * b) = f(a) \circ f(b)$$

Kernel

$$\text{Ker}(f) = \{x \in G: f(x) = e'\}$$

Isomorphism

$$G \cong H$$

25. Lagrange's Theorem

Lagrange's Theorem is one of the most important theorems in Group Theory.

It connects:

- Group
- Subgroup
- Order of group

Statement of Lagrange's Theorem

Theorem: The order of a subgroup divides the order of the group.

Mathematical Representation

$$|H| \mid |G|$$

Where:

Symbol	Meaning
(G)	Group
(H)	Subgroup
$($	G
$($	H
$ $	"divides"

Simple Meaning

Number of subgroup elements
must exactly divide
number of group elements

Important Formula

From Lagrange theorem:

$$|G| = [G] \times |H|$$

Where:

Symbol	Meaning
$([G:H])$	Number of cosets (index of subgroup)

Core Idea Behind Theorem

A group can be partitioned into:

- equal-sized cosets

Each coset has same number of elements as subgroup.

Therefore subgroup size divides group size.

Visual Understanding

Group G

↓ divided into equal cosets

Each coset size = $|H|$

Example 1

Group:

$$G = \mathbb{Z}_6 = \{0,1,2,3,4,5\}$$

Order of group:

$$|G|=6$$

Subgroup:

$$H=\{0,3\}$$

Order of subgroup:

$$|H|=2$$

Check theorem:

$$2 \mid 6$$

True.

Hence verified.

Cosets in Example

Cosets:

Coset	Elements
(0+H)	({0,3})
(1+H)	({1,4})
(2+H)	({2,5})

Each coset has 2 elements.

Example 2

Group:

$$|G|=12$$

Possible subgroup orders are:

Possible?
1
2
3
4
6
12

because all divide 12.

Impossible Orders

Not Possible
5

7
8

because they do not divide 12.

Important Consequences

1. Order of Element Divides Order of Group

If element $a \in G$,
then:

$$o(a) \mid |G|$$

Where:

- $(o(a))$ = order of element

Example

In:

$$\mathbb{Z}_6$$

Group order:6
Possible element orders: 1,2,3,6

2. Prime Order Group Is Cyclic

If:
 $|G|=p$
where (p) is prime,
then group is cyclic.

Example

$|G|=5$
Since 5 is prime:
Group must be cyclic.

3. Subgroup Order Restrictions

Subgroup order must divide group order.
This helps identify valid subgroups quickly.

Example

Can a group of order 10 have subgroup of order 4?
Check:

$$4 \nmid 10$$

No.
Impossible.

Applications of Lagrange Theorem

Area	Usage
Cryptography	Cyclic groups
Coding Theory	Group structures
Abstract Algebra	Subgroup analysis
Computer Science	Finite algebraic systems

Important Notes

Point	Meaning
Applies only to finite groups	✓
Based on cosets	✓
Subgroup order divides group order	✓

Difference Between Group Order and Element Order

Group Order	Element Order
Total group elements	Repetitions needed to get identity

Example of Element Order

In:

$$\mathbb{Z}_4$$

Element: 2

Check: $2+2=0$

Identity obtained after 2 operations.

Therefore: $o(2)=2$

Complete Example

Group:

$$\mathbb{Z}_8$$

Order:

$$|G|=8$$

Possible subgroup orders:

Valid Orders
1
2
4
8

Invalid Orders

Invalid
3
5
6

Summary Table

Concept	Meaning
Lagrange Theorem	Subgroup order divides group order
Cosets	Equal-sized partitions
Index	Number of cosets
Element order	Divides group order

Important GTU Questions

Question Type	Example
State Lagrange theorem	Theory
Verify theorem using example	Numerical
Find possible subgroup orders	Problem solving
Find order of element	Numerical
Applications of theorem	Short note

Most Important Formulas**Lagrange Theorem**

$$|H| \mid |G|$$

Group Order Formula

$$|G| = [G : H] \times |H|$$

Element Order

$$o(a) \mid |G|$$

26. Congruence Relation

Definition: A **Congruence Relation** is a special type of equivalence relation used in modular arithmetic.

Two integers (a) and (b) are congruent modulo (n) if: Their difference is divisible by (n).

Mathematical Representation

$$a \equiv b \pmod{n}$$

Meaning:

$$n \mid (a - b)$$

Divisibility Formula

$$n \mid (a - b)$$

Simple Meaning

a and b leave same remainder when divided by n

Example 1

Check:

$$17 \equiv 5 \pmod{6}$$

Difference:

$$17 - 5 = 12$$

Since:

$$6 \mid 12$$

relation is true.

Example 2

$$20 \equiv 2 \pmod{9}$$

Difference:

$$20 - 2 = 18$$

$$9 \mid 18$$

True.

Example 3

$$11 \equiv 4 \pmod{5}$$

Difference:

$$11 - 4 = 7$$

$$5 \nmid 7$$

False.

Congruence via Remainders

Number	Remainder when divided by 4
1	1
5	1
9	1

Therefore:

$$1 \equiv 5 \equiv 9 \pmod{4}$$

Properties of Congruence Relation

Congruence relation is an equivalence relation.

1. Reflexive

$$a \equiv a \pmod{n}$$

because:

$$a - a = 0$$

and $(n \mid 0)$.

2. Symmetric

If:

$$a \equiv b \pmod{n}$$

then:

$$b \equiv a \pmod{n}$$

3. Transitive

If:

$$a \equiv b \pmod{n}$$

and

$$b \equiv c \pmod{n}$$

then:

$$a \equiv c \pmod{n}$$

Arithmetic Operations

Congruence works with arithmetic operations.

Addition

$$a \equiv b \pmod{n} \Rightarrow a + c \equiv b + c \pmod{n}$$

Multiplication

$$a \equiv b \pmod{n} \Rightarrow ac \equiv bc \pmod{n}$$

Example

$$7 \equiv 2 \pmod{5}$$

Add 3:

$$10 \equiv 5 \pmod{5}$$

Correct.

Important Applications

Area	Usage
------	-------

Cryptography	RSA algorithm
Clock arithmetic	Time systems
Computer Science	Hashing
Coding Theory	Error correction

Clock Arithmetic Example

Clock modulo 12:

$$10 + 5 \equiv 3 \pmod{12}$$

Because after 12, counting restarts.

Congruence Classes

Modulo 3 creates classes:

Class	Elements
([0])	(..., -6, -3, 0, 3, 6, ...)
([1])	(..., -5, -2, 1, 4, 7, ...)
([2])	(..., -4, -1, 2, 5, 8, ...)

Important Notes

Point	Meaning
Congruence based on remainders	✓
Modular arithmetic concept	✓
Forms equivalence classes	✓

27. Quotient Structure

Definition: A **Quotient Structure** is formed by: Dividing an algebraic structure into equivalence classes or cosets.

Simple Meaning

Large structure

↓

Partition into classes

↓

New structure formed

Quotient Group

If:

$$N \trianglelefteq G$$

where (N) is normal subgroup,
then quotient group is:
 G/N

Meaning of (G/N)

Set of all cosets of (N) in (G).

Example 1 — Integers Modulo 3

Take group:

$$(\mathbb{Z}, +)$$

Normal subgroup:

$$3\mathbb{Z}$$

Cosets:

Coset	Elements
$(0 + 3\mathbb{Z})$	$(\dots, -6, -3, 0, 3, 6, \dots)$
$(1 + 3\mathbb{Z})$	$(\dots, -5, -2, 1, 4, 7, \dots)$
$(2 + 3\mathbb{Z})$	$(\dots, -4, -1, 2, 5, 8, \dots)$

These cosets form quotient structure:

$$\frac{\mathbb{Z}}{3\mathbb{Z}}$$

Important Requirement

Quotient group exists only when subgroup is normal.

Condition

$$N \trianglelefteq G$$

Quotient Operation

If operation in (G) is multiplication/addition,
then quotient uses:

$$(aN)(bN) = (ab)N$$

Quotient Formula

$$(aN)(bN) = (ab)N$$

Visual Understanding

Group G

↓

Divide using normal subgroup

↓

Cosets formed

↓

Quotient group

Quotient Structure in Modular Arithmetic

Modulo arithmetic is actually quotient structure.

Example:

$$\mathbb{Z}_n = \frac{\mathbb{Z}}{n\mathbb{Z}}$$

Example

$$\mathbb{Z}_5$$

contains classes:

$[0], [1], [2], [3], [4]$

Important Notes

Point	Meaning
Based on cosets	✓
Requires normal subgroup	✓
Simplifies algebraic structures	✓

Applications

Area	Usage
Abstract Algebra	Group simplification
Coding Theory	Equivalence classes
Cryptography	Modular systems
Computer Science	Finite structures

Difference Between Congruence Relation and Quotient Structure

Congruence Relation	Quotient Structure
Relation between elements	Structure formed from classes
Uses modulo relation	Uses cosets/classes
Equivalence concept	New algebraic system

Summary Table

Topic	Main Idea
Congruence Relation	Same remainder relation
Modular Arithmetic	Arithmetic with remainders
Quotient Structure	Structure of cosets/classes
Quotient Group	(G/N)

Important GTU Questions

Question Type	Example
Define congruence relation	Theory
Verify congruence	Numerical
Find congruence classes	Problem solving
Define quotient group	Theory
Explain (G/N)	Conceptual
Relation between normal subgroup and quotient group	Theory

Most Important Formulas

Congruence Relation

$$a \equiv b \pmod{n}$$

Divisibility Condition

$$n \mid (a - b)$$

Quotient Group

G/N

Quotient Operation

$$(aN)(bN) = (ab)N$$

28. Ring

Definition: A Ring is an algebraic structure having:

- Two binary operations
 - Addition (+)
 - Multiplication ($\times$)
- Special properties for both operations.

Structure Representation

$$(R, +, \times)$$

Where:

- (R) = set
- $(+)$ = addition
- $(\times)$ = *multiplication*

Conditions of Ring

A set (R) forms a ring if:

Under Addition $((+))$

The set forms an Abelian Group.

Meaning:

Property	Required
Closure	✓
Associative	✓
Identity (0)	✓
Inverse	✓
Commutative	✓

Under Multiplication $((\times))$

Property	Required
Closure	✓
Associative	✓

Multiplication inverse is NOT compulsory.

Distributive Laws

Multiplication distributes over addition.

Left Distributive Law

$$a(b+c)=ab+ac$$

Right Distributive Law

$$(a+b)c=ac+bc$$

Simple Understanding

Addition behaves like Abelian group +

Multiplication behaves associative =

Ring

Example 1 — Integers

$$(\mathbb{Z}, +, \times)$$

Check addition:

Property	Result
Closure	✓
Associative	✓
Identity	0
Inverse	-a
Commutative	✓

Check multiplication:

Property	Result
----------	--------

Closure	✓
Associative	✓

Distributive laws hold.

Therefore integers form a ring.

Example 2 — Even Integers

$$(2\mathbb{Z}, +, \times)$$

Also forms a ring.

Important Notes

Point	Meaning
Two operations exist	✓
Addition part stronger than multiplication	✓
Multiplication inverse not required	✓

Types of Rings

Type	Property
Commutative Ring	Multiplication commutative
Ring with Unity	Multiplicative identity exists
Integral Domain	No zero divisors
Field	Every nonzero element invertible

Ring With Unity

If multiplication identity exists:

$$a \cdot 1 = 1 \cdot a = a$$

then ring is called ring with unity.

Example

Integers:

$$(\mathbb{Z}, +, \times)$$

have multiplicative identity: 1

Hence ring with unity.

Non-Example

Natural numbers under usual operations:

$$(\mathbb{N}, +, \times)$$

No additive inverse exists.

Therefore not ring.

Real-Life Analogy

Addition system +

Multiplication system

working together

29. Commutative Ring

Definition : A **Commutative Ring** is a ring in which: Multiplication is commutative.

Commutative Multiplication Formula

$$ab=ba$$

Conditions

A commutative ring satisfies:

1. All ring properties
2. Multiplication commutativity

Simple Understanding

Ring +

$ab = ba =$

Commutative Ring

Example 1 — Integers

$$(\mathbb{Z}, +, \times)$$

Because:

$ab=ba$

Integers form commutative ring.

Example 2 — Real Numbers

$$(\mathbb{R}, +, \times)$$

Multiplication commutative.

Hence commutative ring.

Example of Non-Commutative Ring

Matrix ring:

$$M_n(\mathbb{R})$$

because generally:

$$AB \neq BA$$

Therefore not commutative ring.

Important Notes

Point	Meaning
Every commutative ring is ring	✓
Every ring need not be commutative	✓
Matrix multiplication common counterexample	✓

Ring Hierarchy

Ring

↓

Commutative Ring

↓

Integral Domain

↓

Field

Comparison Table

Property	Ring	Commutative Ring
Two operations	✓	✓
Addition Abelian group	✓	✓
Multiplication associative	✓	✓
Multiplication commutative	✗ not necessary	✓

Important Ring Properties

Additive Identity

$$a+0=a$$

Additive Inverse

$$a+(-a)=0$$

Multiplication Associative

$$(ab)c=a(bc)$$

Distributive Law

$$a(b+c)=ab+ac$$

Example Verification

Check whether:

$$(\mathbb{Z}, +, \times)$$

is commutative ring.

Property	Result
Addition Abelian group	✓
Multiplication associative	✓
Multiplication commutative	✓
Distributive law	✓

Therefore commutative ring.

Applications of Rings

Area	Usage
Cryptography	RSA algorithms
Coding Theory	Error correction
Computer Science	Abstract computation
Digital Systems	Algebraic logic

Most Important Formulas

Left Distributive Law

$$a(b+c)=ab+ac$$

Right Distributive Law

$$(a+b)c=ac+bc$$

Commutative Multiplication

$$ab=ba$$

30. Integral Domain

Definition: An **Integral Domain** is a: Commutative ring with unity having no zero divisors.

Structure Requirements

To become an integral domain, a structure must satisfy:

1. Ring properties
2. Multiplication commutative
3. Multiplicative identity exists
4. No zero divisors

Mathematical Representation

$$(D, +, \times)$$

Core Property

No Zero Divisors

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0$$

Meaning

If multiplication gives zero:

- at least one factor must be zero.

Simple Understanding

Nonzero $\times$ Nonzero

can never become zero

What Are Zero Divisors?

A nonzero element (a) is called zero divisor if:

$$ab=0$$

for some nonzero (b).

Example of Zero Divisors

Consider:

$$\mathbb{Z}_6 = \{0,1,2,3,4,5\}$$

Check:

$$2 \times 3 = 6 \equiv 0 \pmod{6}$$

But:

$$2 \neq 0, 3 \neq 0$$

Hence:

- 2 and 3 are zero divisors.

Therefore:

$$\mathbb{Z}_6$$

is NOT integral domain.

Example of Integral Domain Integers

$$(\mathbb{Z}, +, \times)$$

Check:

Property	Result
Commutative ring	✓
Unity exists (1)	✓
No zero divisors	✓

Therefore integers form integral domain.

Example 2 — Real Numbers

$$(\mathbb{R}, +, \times)$$

No zero divisors exist.

Hence integral domain.

Important Notes

Point	Meaning
Every integral domain is commutative ring	✓
Zero divisors not allowed	✓
Multiplicative identity compulsory	✓

Non-Examples

Structure	Reason
$(\mathbb{Z}_6)$	Zero divisors exist
Matrix rings	Multiplication not commutative

Important Property

Cancellation law holds in integral domains.

Cancellation Law

$$ab = ac, a \neq 0 \Rightarrow b = c$$

Why?

Because no zero divisors exist.

Example

In integers:

$$2x = 2y$$

Then:

$$x = y$$

Hierarchy

Ring

↓

Commutative Ring

↓

Integral Domain

31. Field

Definition

A **Field** is an integral domain in which: Every nonzero element has multiplicative inverse.

Main Requirement

For every nonzero (a):

$$a \cdot a^{\{-1\}} = 1$$

Structure Requirements

A field satisfies:

1. Integral domain properties
2. Every nonzero element invertible

Simple Understanding

Addition works

Multiplication works

Division also works

(except divide by zero)

Example 1 — Rational Numbers

$$(\mathbb{Q}, +, \times)$$

Check:

Property	Result
Integral domain	✓
Every nonzero element invertible	✓

Example:

$$2^{-1} = \frac{1}{2}$$

Hence field.

Example 2 — Real Numbers

$$(\mathbb{R}, +, \times)$$

Every nonzero element has inverse.
Therefore field.

Example 3 — Complex Numbers

$$(\mathbb{C}, +, \times)$$

Also field.

Non-Example — Integers

$$(\mathbb{Z}, +, \times)$$

Inverse of: 2
is: $\frac{1}{2}$
But:

$$\frac{1}{2} \notin \mathbb{Z}$$

Therefore integers are NOT field.

Important Notes

Point	Meaning
Every field is integral domain	✓
Division possible	✓
Every nonzero element invertible	✓

Multiplicative Inverse Examples

Element	Inverse
2	$\left(\frac{1}{2}\right)$
5	$\left(\frac{1}{5}\right)$
(-3)	$\left(-\frac{1}{3}\right)$

Finite Field Example

$$\mathbb{Z}_5$$

forms field because 5 is prime.

Important Rule

$\mathbb{Z}_p$

is field if (p) is prime.

Example

$\mathbb{Z}_7$

Field ✓
because 7 is prime.

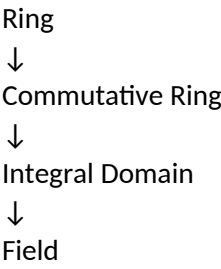
Counterexample

$\mathbb{Z}_6$

Not field ✗
because:

- zero divisors exist
- some inverses absent

Field Hierarchy



Comparison Table

Property	Ring	Integral Domain	Field
Commutative multiplication	Optional	✓	✓
Unity exists	Optional	✓	✓
No zero divisors	✗	✓	✓
Every nonzero element invertible	✗	✗	✓

Quick Example Table

Structure	Ring	Integral Domain	Field
($\mathbb{Z}$)	✓	✓	✗
($\mathbb{Q}$)	✓	✓	✓
($\mathbb{R}$)	✓	✓	✓
($\mathbb{Z}_6$)	✓	✗	✗
($\mathbb{Z}_5$)	✓	✓	✓

Real-Life Analogy

Structure	Analogy
Ring	Addition + multiplication
Integral Domain	Safe multiplication
Field	Full arithmetic system

Applications

Area	Usage
------	-------

Cryptography	Finite fields
Coding Theory	Error correction
Computer Graphics	Coordinate systems
AI & ML	Linear algebra

Important GTU Questions

Question Type	Example
Define integral domain	Theory
Define field	Theory
Explain zero divisor	Conceptual
Difference between ring and field	Theory
Check whether structure is field	Numerical
Explain cancellation law	Problem solving

Most Important Formulas

No Zero Divisors

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0$$

Multiplicative Inverse

$$a \cdot a^{\{-1\}} = 1$$

Cancellation Law

$$ab = ac, a \neq 0 \Rightarrow b = c$$